Clase 3 Fundamentos del Aprendizaje profundo

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Definiciones

- Muestra o entrada
- Predicción o salida
- Error de predicción o valor de perdida
- Clases
- Etiqueta
- Clasificación binaria
- Clasificación categórica o clasificación multiclase
- Clasificación multietiqueta
- Regresión escalar
- Regresión vector
- Mini lote o lote

Reflexiones de los colabs

- Las tareas más comunes del aprendizaje de maquina son: clasificación binaria, clasificación multiclase y regresión escalar.
- Cada una usa diferentes funciones de perdida: binary_crossentropy, categorical_crossentropy y mean_squared_error.
- Se necesita preprocesar los datos crudos antes de cargarlos a la red neuronal.
- Normalizar las características si existen rangos.
- Pequeños modelos cuando los datos son pequeños.
- No causar cuellos de botella dependiendo de la última capa.
- Con pocos datos usar validación cruzada.

One-hot, TF-IDF y embeddings

¿QUÉ ES UN VECTOR?

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Figure 6.2 The term-document matrix for four words in four Shakespeare plays. Each cell contains the number of times the (row) word occurs in the (column) document.

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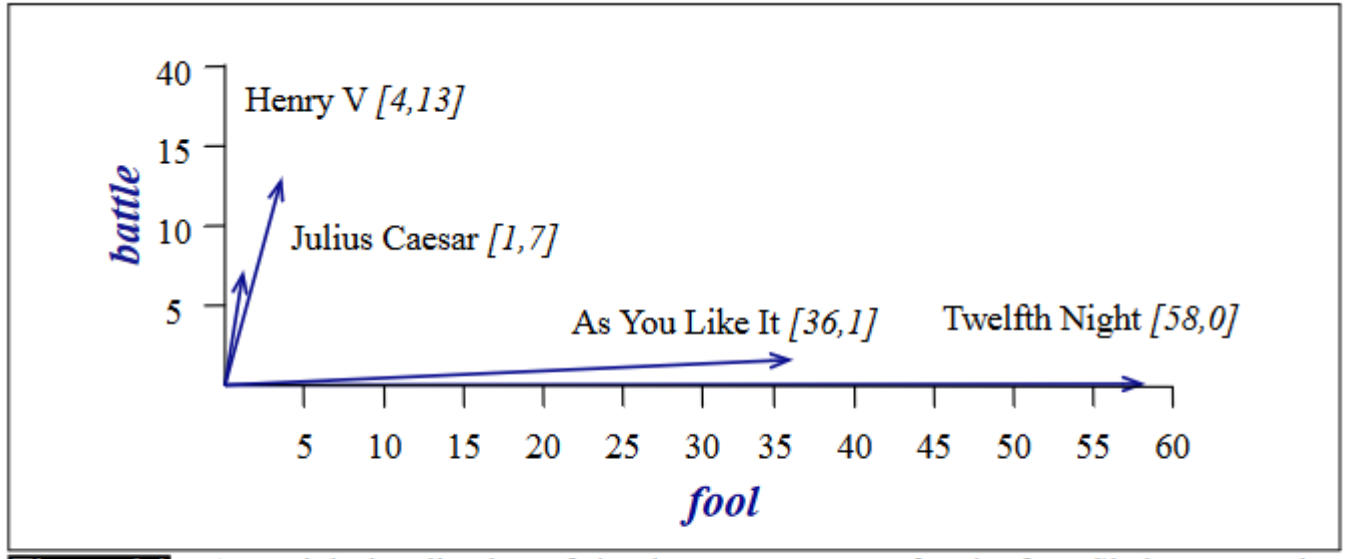


Figure 6.4 A spatial visualization of the document vectors for the four Shakespeare play documents, showing just two of the dimensions, corresponding to the words *battle* and *fool*. The comedies have high values for the *fool* dimension and low values for the *battle* dimension.

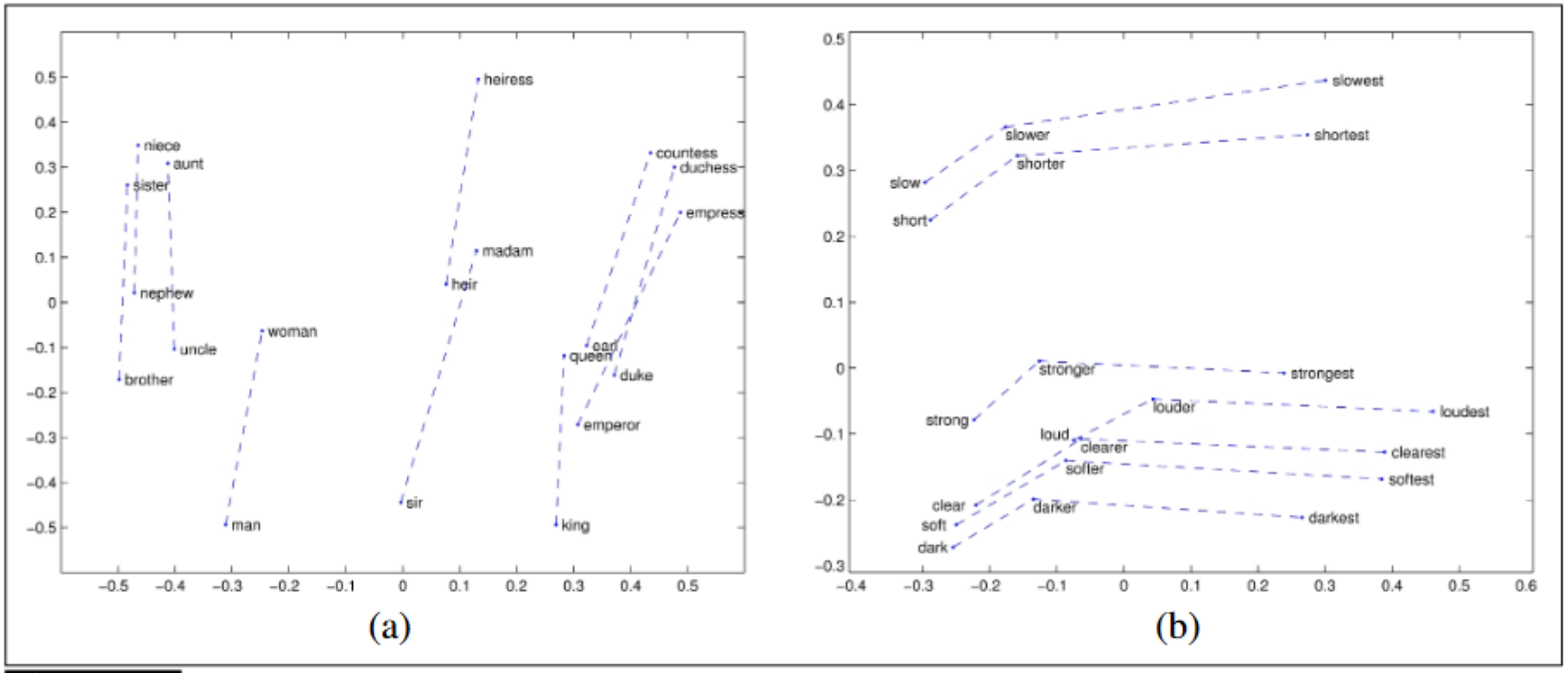


Figure 6.16 Relational properties of the GloVe vector space, shown by projecting vectors onto two dimensions. (a) $\overrightarrow{(\text{king})} - \overrightarrow{\text{man}} + \overrightarrow{\text{woman}}$ is close to $\overrightarrow{\text{queen}}$. (b) offsets seem to capture comparative and superlative morphology (Pennington et al., 2014).

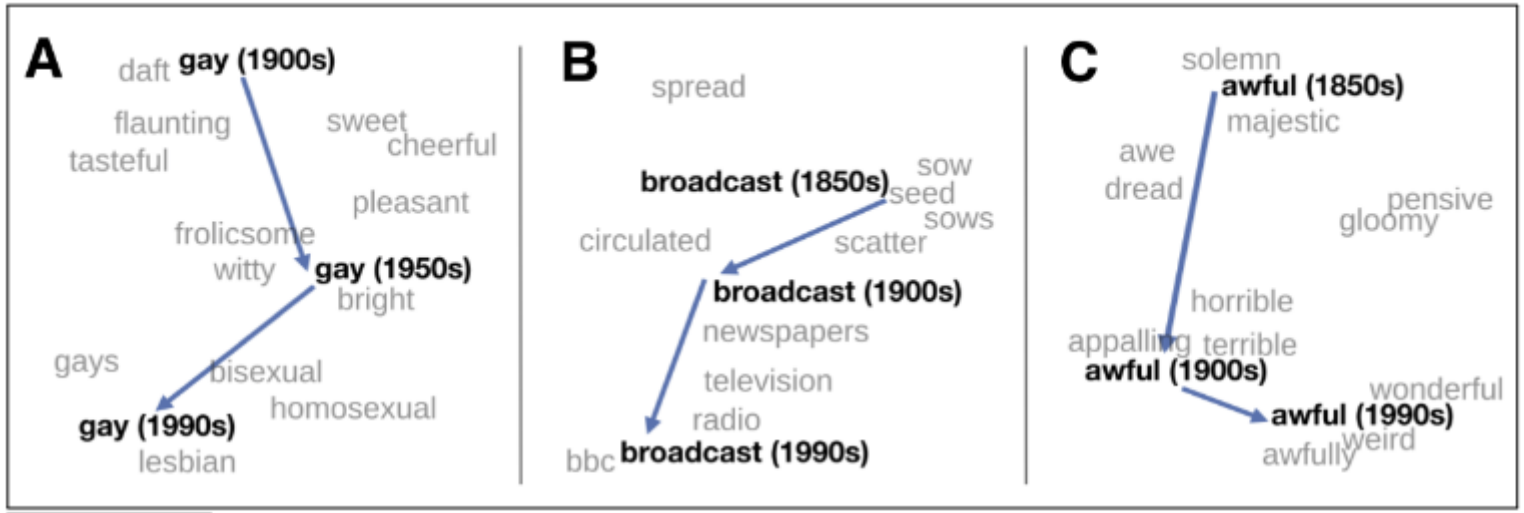
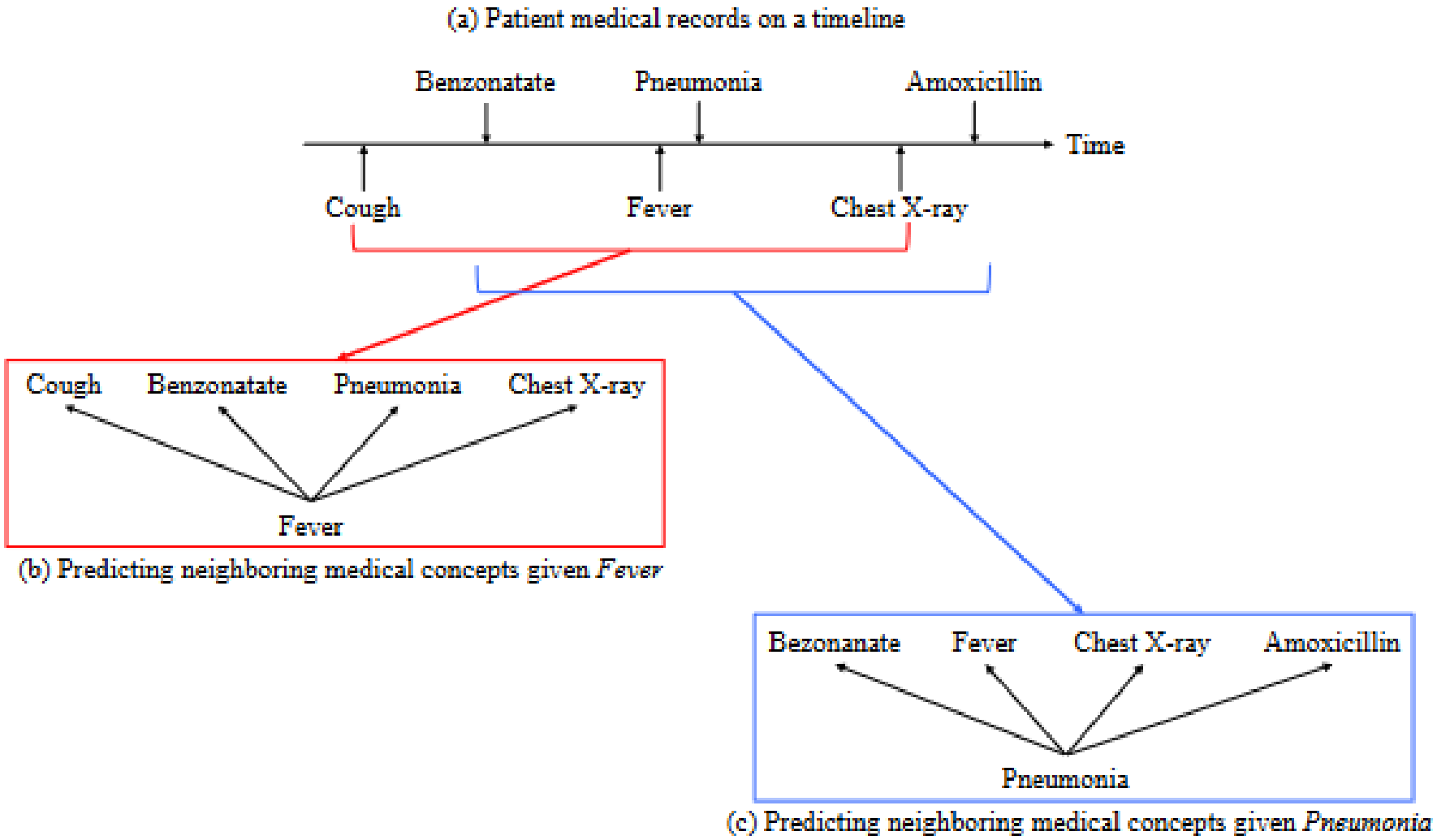


Figure 6.17 A t-SNE visualization of the semantic change of 3 words in English using word2vec vectors. The modern sense of each word, and the grey context words, are computed from the most recent (modern) time-point embedding space. Earlier points are computed from earlier historical embedding spaces. The visualizations show the changes in the word *gay* from meanings related to "cheerful" or "frolicsome" to referring to homosexuality, the development of the modern "transmission" sense of *broadcast* from its original sense of sowing seeds, and the pejoration of the word *awful* as it shifted from meaning "full of awe" to meaning "terrible or appalling" (Hamilton et al., 2016b).

WORD2VEC CLINICO



WORD2VEC CLINICO

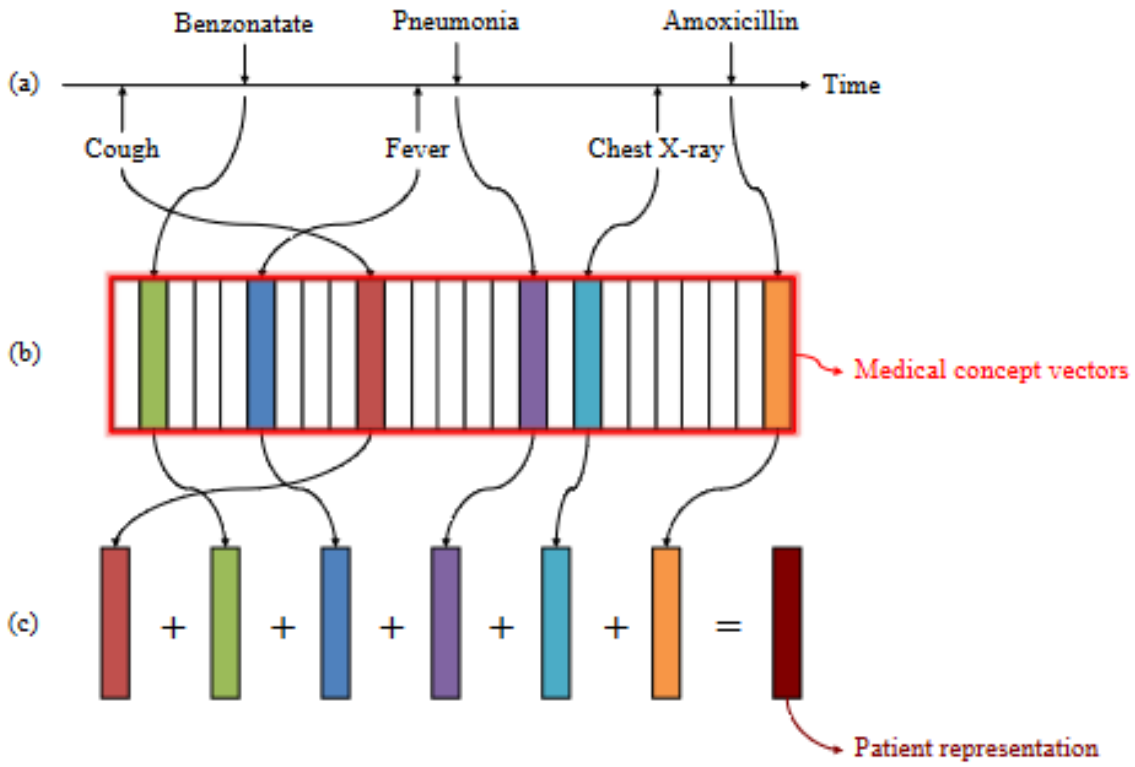


Figure 4. Patient representation construction. (a) represents a medical record of a patient on a timeline. (b) The medical concepts are represented as vectors using the trained medical concept vectors. (c) The patient is represented as a vector by summing all medical concept vectors.

TF: term frequency (frecuencia del término)

$$tf_{t,d} = count(t,d)$$
$$tf_{t,d} = log_{10}(count(t,d) + 1)$$

IDF: inverse document frequency (frecuencia invertida del documento)

$$idf_t = log_{10}(\frac{N}{df_t})$$

Valor ponderado tf-idf para una palabra t

$$w_{t,d} = tf_{t,d} * idf_t$$

EJEMPLO EN EXCEL

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
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Extracción de características

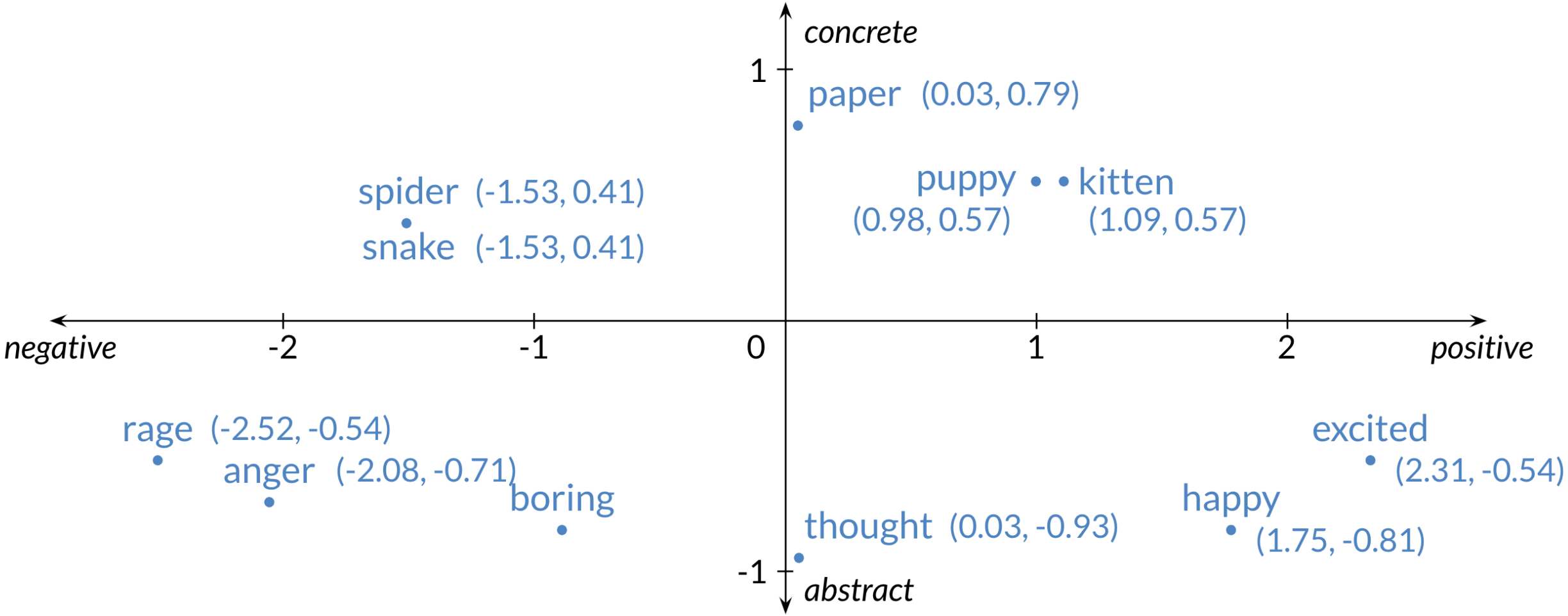
Tf-idf (Term frequency – Inverse document frequency), frecuencia de término – frecuencia inversa de los documentos en el cual ocurre el término.

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	0.074	0	0.22	0.28
good	0	0	0	0
fool	0.019	0.021	0.0036	0.0083
wit	0.049	0.044	0.018	0.022

Figure 6.9 A tf-idf weighted term-document matrix for four words in four Shakespeare plays, using the counts in Fig. 6.2. For example the 0.049 value for *wit* in *As You Like It* is the product of $\text{tf} = \log_{10}(20+1) = 1.322$ and $\text{idf} = .037$. Note that the idf weighting has eliminated the importance of the ubiquitous word *good* and vastly reduced the impact of the almost-ubiquitous word *fool*.

CBOW

SKIP-GRAM


concrete
1
paper (0.03, 0.79)
puppy
kitten
(0.98, 0.57)
(1.09, 0.57)
spider (-1.53, 0.41)
snake (-1.53, 0.41)
negative
-2
-1
0
1
2
positive
rage (-2.52, -0.54)
anger (-2.08, -0.71)
boring
excited
(2.31, -0.54)
happy
(1.75, -0.81)
thought (0.03, -0.93)
-1
abstract

Optimización vs generalización

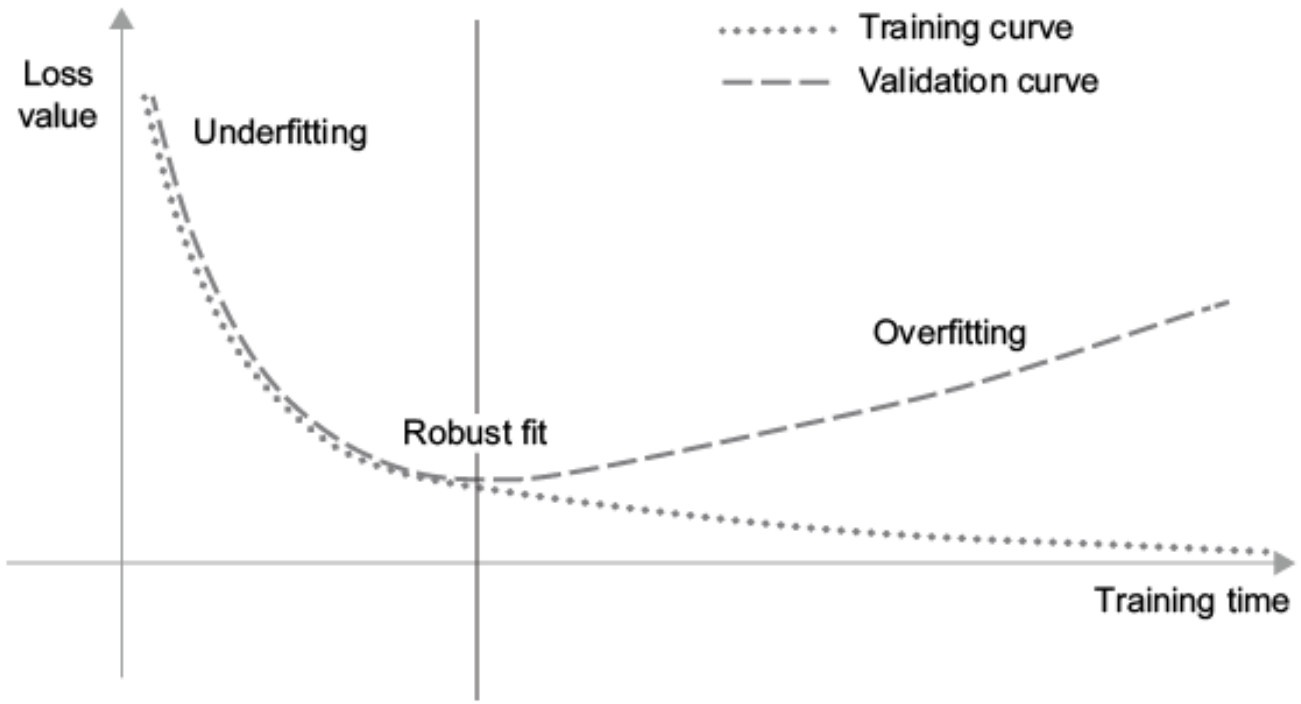


Figure 5.1 Canonical overfitting behavior

Ruido

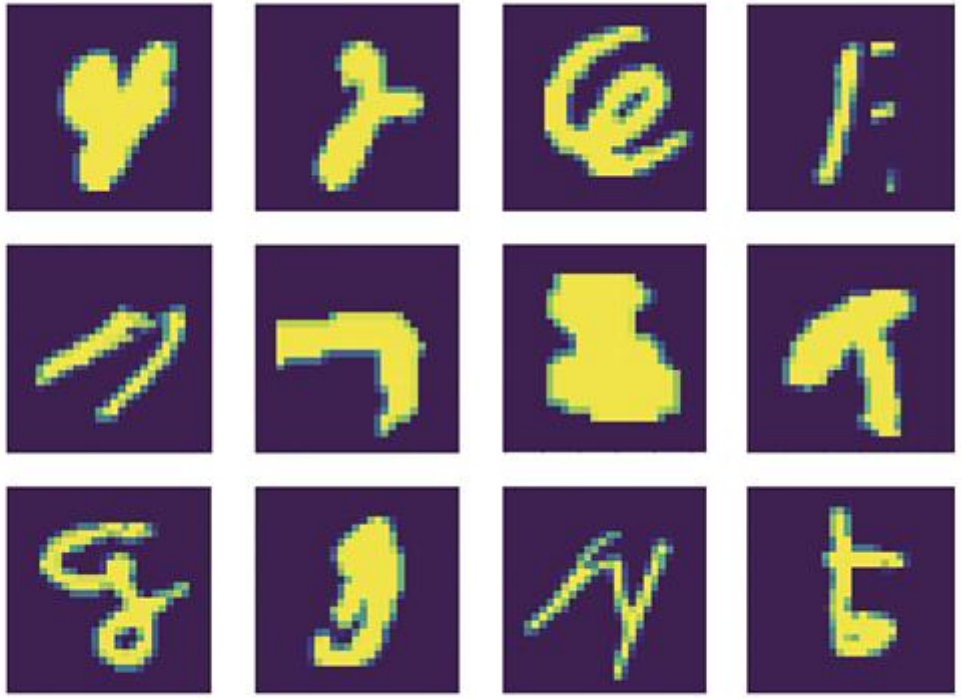


Figure 5.2 Some pretty weird MNIST training samples

Etiquetas

Label: 9 Label: 7 Label: 4 Label: 3 Label: 5

Figure 5.3 Mislabeled MNIST training samples

La naturaleza de la generalización

- No todo ruido proviene de características ambiguas.
- Ponerse de acuerdo en un contexto local, no implica que se resuelva la ambigüedad.
- La hipótesis de la variedad topológica.
- El reconocimiento de patrones se puede hacer si se cumple la variedad topológica.
- Los modelos de aprendizaje profundo implementan un suave y continuo mapeo de los datos de entrada a la salida.

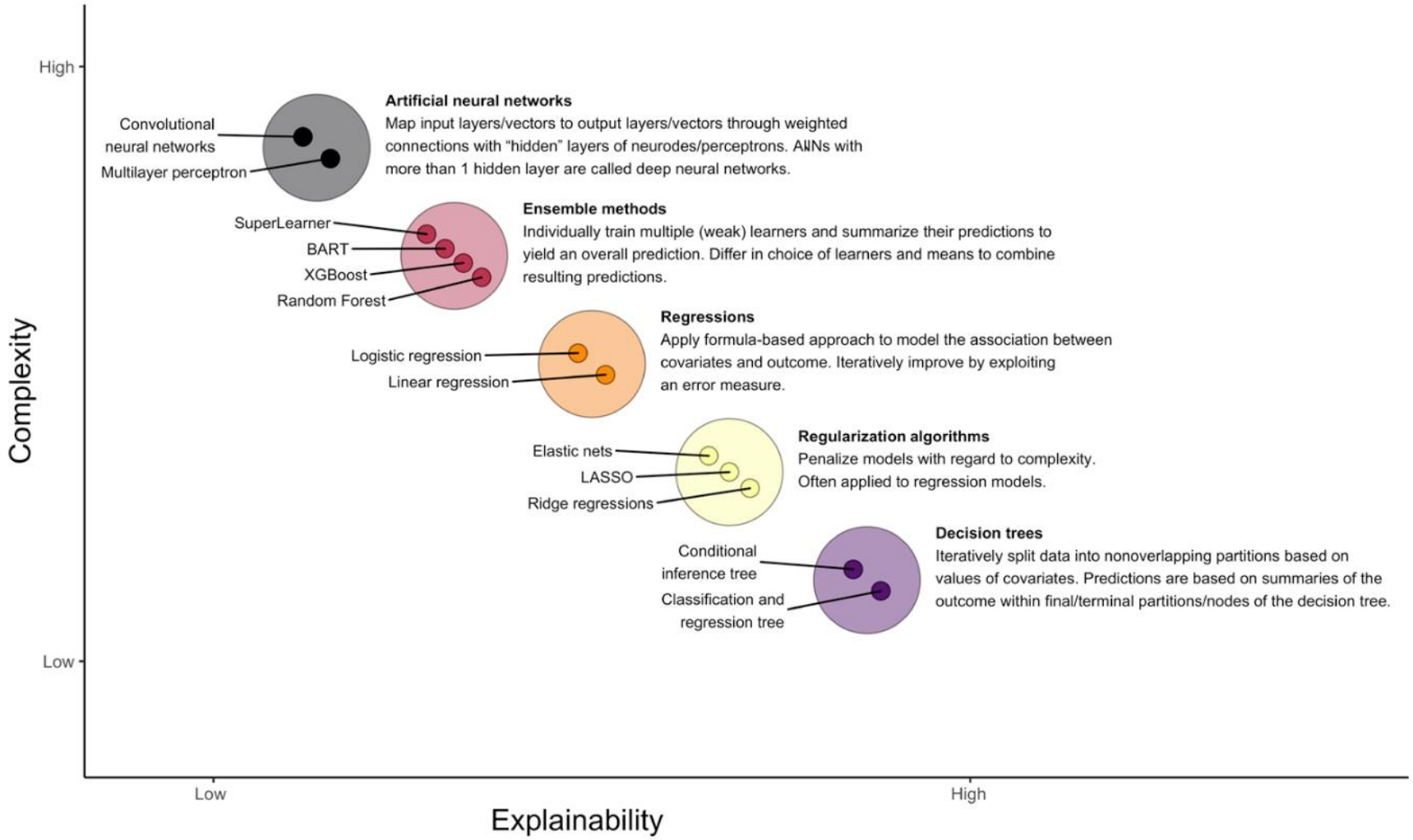


Fig. 2. ML methods for prediction most relevant in the social and health sciences with nontechnical description ranked by interpretability/explainability versus complexity. Note that classes of methods are represented as larger circles; specific ML methods are represented as small circles within. Ordering and selection of ML methods based on theoretical considerations and experience. ANN, artificial neural network. Copyright by Matthias Klee.